# Kuhn–Tucker in pictures

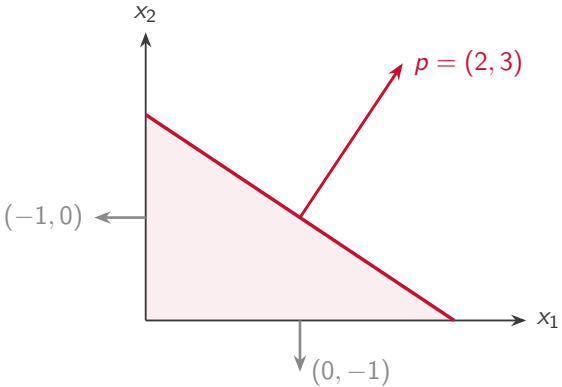
A note to class 4 · Microeconomics I

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also called KKT: the Karush–Kuhn–Tucker conditions

# Three fences around the budget set




$\max_x \; u(x)$ subject to

$p \cdot x \leq w$

$x_1 \geq 0, \;\; x_2 \geq 0$

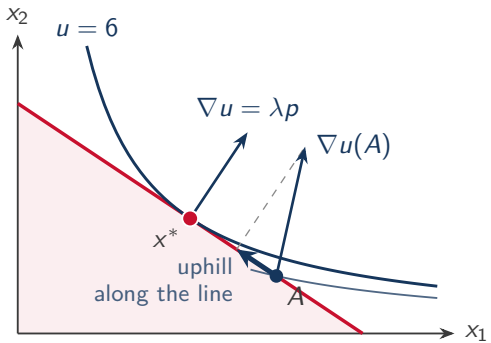
written as $-x_1 \leq 0, \;\; -x_2 \leq 0$

arrows: the constraints' gradients

Each constraint's gradient points where that fence forbids you to go.

$p = (2, 3)$, $w = 12$, as in class 4; MWG 3.D, pp. 53–55.

# On the budget line, $\nabla u$ must point straight through the fence




$u = x_1 x_2$

$A = (4.5, 1)$: $\nabla u = (1,\ 4.5)$

not along $p$: a part points along the line

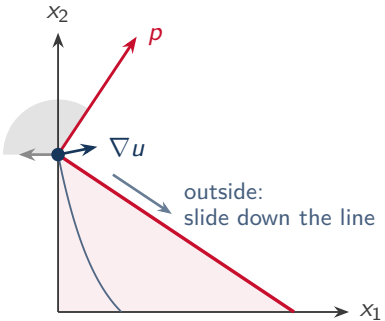
$x^* = (3, 2)$: $\nabla u = (2, 3) = \lambda p$

with $\lambda = 1$: no part along the line

$\nabla u = \lambda p$ with $\lambda \geq 0$: a fence can push back, never pull.

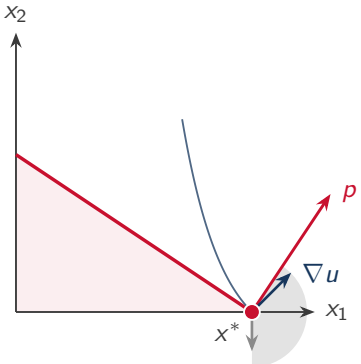
Class 4, slides 21 and 25; gradients drawn at half length.

# At a corner, $\nabla u$ must lie in the cone of the fences you touch




**$(0, 4)$: not optimal**

$\nabla u = (1, \frac{1}{5})$ lies outside the cone



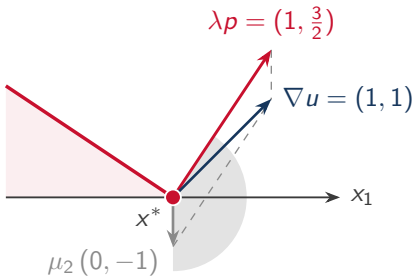

$x^* = (6, 0)$

$\nabla u = (1, 1)$ lies inside the cone

The cone: all $\lambda p + \mu_\ell \cdot$(fence normal) with $\lambda, \mu_\ell \geq 0$.

$u = x_1 + \ln(1 + x_2)$: class 3, slide 27; class 4, slides 23 and 25.

# The multipliers are the push of each fence




$$(1, 1) = \tfrac{1}{2}(2, 3) + \tfrac{1}{2}(0, -1)$$

- $\lambda = \frac{1}{2}$: one more euro, $\frac{1}{2}$ more utility
  $x(p, w) = (w/2,\ 0)$, so $v = w/2$
- $\mu_2 = \frac{1}{2}$: a forced unit of good 2 costs 3 euros = 1.5 units of good 1: $-\frac{3}{2} + 1 = -\frac{1}{2}$
- at $(0, 4)$: $\mu_1 = -\frac{13}{15} < 0$: the fence would have to pull
  $(1, \frac{1}{5}) = \frac{1}{15}(2, 3) - \frac{13}{15}(-1, 0)$

$u = x_1 + \ln(1 + x_2)$, $p = (2, 3)$, $w = 12$; utility changes at the margin.

## From the picture to the conditions

$\nabla u(x^*) = \lambda p - \mu, \qquad \lambda \geq 0, \;\; \mu \geq 0$

$-\mu = \mu_1 (-1, 0) + \mu_2 (0, -1)$: the fences' pushes of slide 5

$\mu_\ell > 0$ only if $x^*_\ell = 0$; $\quad \lambda > 0$ only if $p \cdot x^* = w$

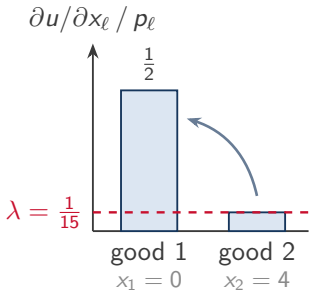
complementary slackness

eliminate $\mu$: $\dfrac{\partial u(x^*)}{\partial x_\ell} \leq \lambda p_\ell$, with equality if $x^*_\ell > 0$ (3.D.1)

Lagrangian $L = u(x) + \lambda (w - p \cdot x) + \mu \cdot x$; set $\partial L / \partial x = 0$.

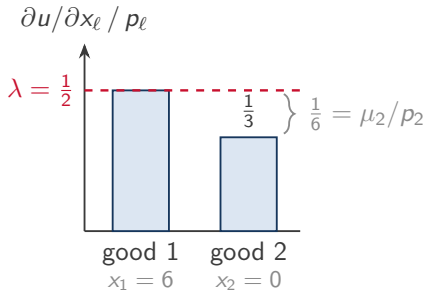
In vectors: $\nabla u(x^*) \leq \lambda p$ and $x^* \cdot [\nabla u(x^*) - \lambda p] = 0$, MWG (3.D.2)–(3.D.3), 3.D, p. 53.

# In euros: every euro goes where it buys the most utility




**at $(0,4)$: not optimal**

good 1 gives $\frac{1}{2} > \lambda$: move euros to it



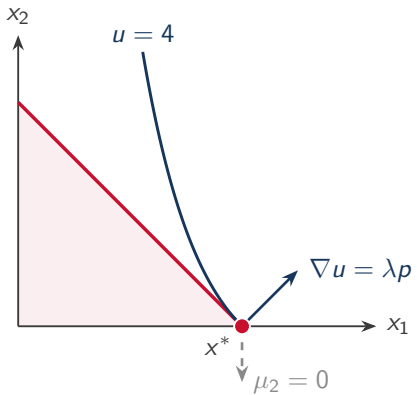

**at $x^* = (6,0)$**

good 2 gives only $\frac{1}{3} < \lambda$: not bought

Bought goods give exactly $\lambda$ per euro; goods not bought give at most $\lambda$.

$u = x_1 + \ln(1 + x_2)$, $p = (2,3)$, $w = 12$; at $(0,4)$ only good 2 is bought, so $\lambda$ would be $\frac{1}{15}$.

## Zero and equality at once




$u = x_1 + \ln(1 + x_2)$

$p = (1, 1)$, $w = 4$: class 4, slide 28, row 3

$x^* = (4, 0), \quad \lambda = 1$

$\partial u / \partial x_2 = 1 = \lambda p_2$

so $\mu_2 = 0$, although $x_2^* = 0$

"Equality if $x_\ell > 0$" means if, not only if.

The indifference curve $x_1 = 4 - \ln(1 + x_2)$ is tangent to the budget line at the corner.

## The general form: any number of fences

$\max_x \; f(x)$ subject to $g_k(x) \le c_k, \;\; k = 1, \ldots, K$

if $x^*$ solves it and the gradients of the binding constraints are linearly independent, then

$$\nabla f(x^*) = \textstyle\sum_k \lambda_k \nabla g_k(x^*), \qquad \lambda_k \ge 0, \qquad \lambda_k \left[c_k - g_k(x^*)\right] = 0$$

consumer: $f = u$, $g = (p \cdot x, \; -x_1, \; -x_2)$, $c = (w, 0, 0)$, multipliers $(\lambda, \mu_1, \mu_2)$

$f$ and $g_k$ continuously differentiable; MWG Mathematical Appendix M.K. In the consumer's problem the qualification always holds: MWG 3.D, footnote 8, p. 53.

## A recipe for the consumer's problem

1. try interior: $\nabla u = \lambda p, \;\; p \cdot x = w$
2. a negative $x_\ell$? set it to 0, solve for the rest
3. check $\lambda \geq 0$ and $\partial u / \partial x_\ell \leq \lambda p_\ell$ where $x_\ell = 0$
4. $u$ quasiconcave, monotone, $\nabla u \neq 0$: the conditions are also sufficient

Sufficiency: MWG 3.D, p. 55. Class 4, slide 28 runs the recipe on three cases.